Native oracle and presentation components

2026-10-18: audit items 4, 15, 16, 17

Status: partial against the original requests. All fixtures are public and synthetic. Complete four-presentation descent, measured solving degree, production integration and full n83 relation solving remain open. Input polynomial degree is different from F4 solving degree.

Host: shared Linux x86_64, four-core CPU quota, 16GiB memory, L0. No timing comparison or speedup is admitted. Frozen protocols, commands, source/binary hashes, raw records, source snapshots and failures are retained.

Multiword n83 construction

Modulus `0x80000000000000000095`; primitive 167th root `0x76e02ca1a43987ce0230e`. C1 has 167 elements; C2 has 13,861 distinct sums of two different roots, with every witness retained. The ONB has rank 83; both sets are Frobenius closed. All 334 generic-backend checks and 4096 full-width probes agreed; all probes were nonmembers. Generic replay checked all 13,861 witnesses and Frobenius images, plus 167 root-order steps. An altered witness was rejected with exit101; the original passed.

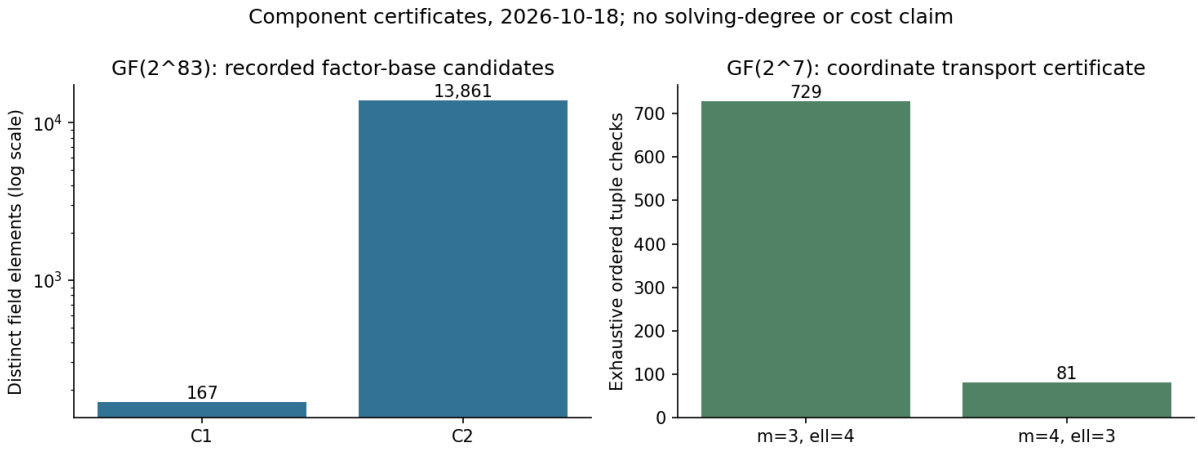
This component needs no exponential subgroup-log table. It does not perform subgroup projection, useful-rank measurement or a full n83 solve. Existing production cyclotomic code remains single-word.

Coordinate and complete addition strata

Curve `icv1-f2m7-t13-616700dd`: K0, modulus 0x83, order 116. The map $[X : Z] \mapsto [Z : X + Z]$ retains its pole and infinity; exact polynomial-prefix bases are transported unchanged.

(m, ℓ)	Lifted points	Pair checks	Tuple checks	Pole points	Distinct sums
$(3, 4)$	9	100	729	2	56
$(4, 3)$	3	16	81	2	4

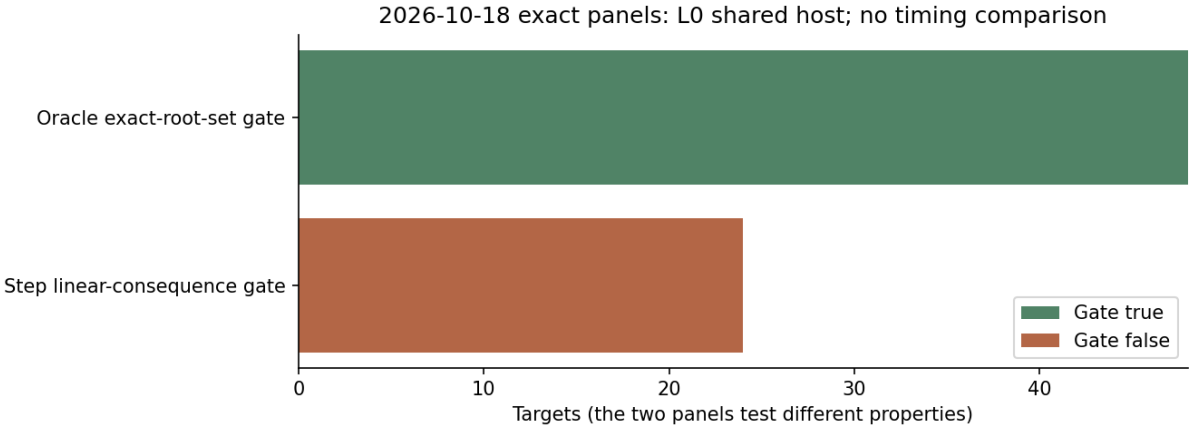
Identity, inverse/two-torsion, doubling and generic checks passed. Every tuple retained its sum. The ell=3 base has only three small-torsion points; it supplies no effectiveness or yield evidence.



Reproduction. Study directory: `research/ideas_execution_20261018/oracle_integration/`. See `README.md`, `commands.txt`, `status.json`, the frozen protocols and `runs/`. Native source: `examples/ideas_oracle_components.rs` at repository root.

Exact original equivariant panels

All 24 step targets($n = 7, 11, 13$,8 each) and 48 full-oracle targets ($n = 5, 7, 11, 13$,12 each) ran. Step tests equality of refutation and linear consequences; oracle checks exact roots against independent brute enumeration.



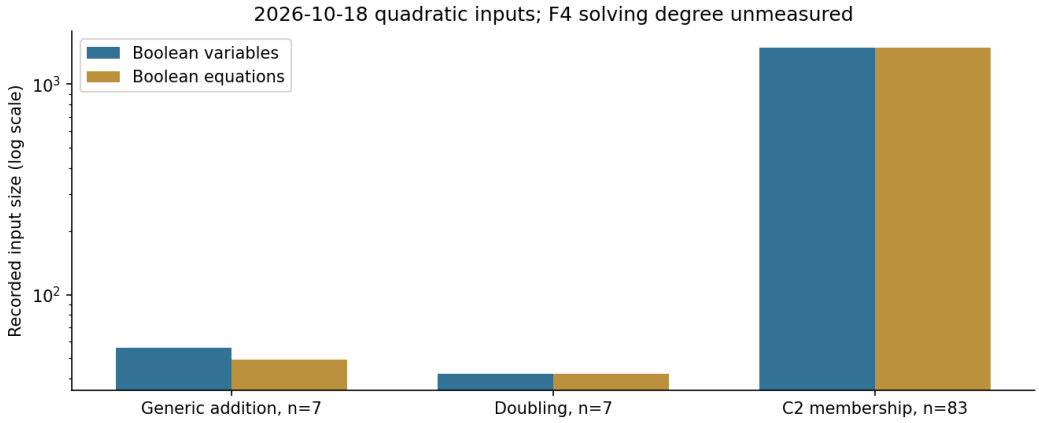
The step replacement gate failed 24/24; containment held 24/24. The separate full-oracle gate passed 48/48, including the repeated incumbent. Fixed-target production lacks the barrel system's Frobenius symmetry. No invalid production hook was installed.

Executed quadratic Boolean inputs

Input	Variables	Equations	Monomials	Input degree
Generic addition(n7)	56	49	952	2
Doubling(n7)	42	42	580	2
C2 membership(n83)	1494	1494	159322	2

Addition charts include nonzero-denominator inverse witnesses. Curve ANFs agreed with independent membership on all 16,384 affine coordinate pairs. Graph zero sets matched independent sums on all 8280 and 690 applicable pair/output assignments over all 115 finite curve points. Exceptional strata retain separate certificates.

The n83 input encodes $z_0^{167} = z_1^{167} = 1$, $x = z_0 + z_1$, $xw = 1$ with Frobenius linear wires and quadratic products. All 32 positives passed; all 32 changed-x and 32 repeated-root controls were rejected. Complete ANFs are retained; F4 on 1494 variables has not been run.



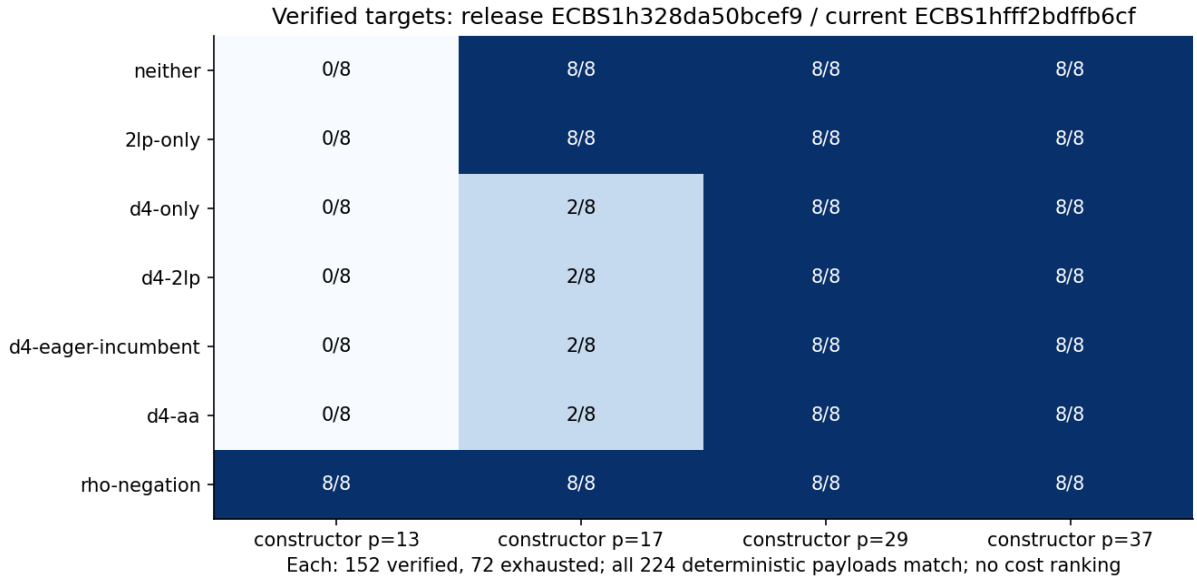
Remaining original criteria: Edwards/theta equations, complete same-base four-presentation m-sum systems, measured D/Macaulay/solve costs, production ecbench ablations, n83 subgroup/decomposition continuation and useful rank. Root's n23 solver package is separate.

Additional complete factorial attempts

The unchanged seven-arm spec first ran under the owner's published native release v0.1.2538. Its source commit is:

`b1bf5b8935f68257dfdd3fc542cb2fef41db741c`

The archive passed publisher checksums. Session `ECBS1h328da50bcef9` completed all 224 attempts: 152 verified, 72 exhausted. A/A matched 32/32 workloads, including failures. Audit passed 152 verified-run replays with zero problems; exhausted records were retained but not replayed. The prior 36-record D4 session also audited cleanly with 12 exact replays.



Admission limits. Four constructor curves await registration. IC charges include a historical algebraic-solve model and unpriced stand-in oracle. Native comparisons mark the panel incomplete and diagnostic. The 0/0.1/1 GAE band covers every declared unpriced counter on every arm, including rho; missing counters are exposed. Unknown non-emitted bookkeeping remains outside that band. No complete-cost ranking or citable curve-specific conclusion is admitted. Native output permits wall values under its L0 spec; this report rejects that timing admission.

Current-build repair. The CLI initially failed because of a missing `fbm` import. Root authorized the minimal import; current CLI test and release build passed. Current session `ECBS1hfff2bdffb6cf` repeated all 224 attempts with the same 152 verified/72 exhausted outcomes, 32/32 A/A matches, and 152 verified-run replays. An exact native comparison matched all 224 deterministic target/outcome/cost/counter/detail payloads across the two binaries, including all 72 exhausted runs. Timings/source labels were excluded. The release session was not overwritten.

Prior evidence retained. D4+2LP had already measured 1.025 [1.016,1.037] on 24 held-out curves; round2 eager D4 measured 0.9792 [0.9725,0.9855] on 27. The trace filter rejected 0/29,624 subgroup targets. Free relations are already production default. These are prior results.

Validation. Component Cargo tests passed 3/3; release build passed. The analysis example built and its zero-test target passed. Parent's full library suite reported 4378 passed, 4 failed, 117 ignored before separate baseline repair. Those files were not edited here; no full-suite pass is claimed.

Sources. Study `commands.txt` and `runs/`; prior studies `research/ic_improve_20261014/01-d4-2lp/`, `research/ic_improve_20261015/04-d4-2lp-fix/`, `research/ic_improve_20261014/03-tracezero-freerel/`, and `research/ic_improve_20261015/02-freerel-production/`.